MATH INQUIRERS PLACEMENT CHECK

Ages 8-10

Typically children finishing Primary 3, and open to a year either side • About 30 minutes • No calculator

Name: _____ Date: _____

How it works. Your child answers on their own. Read a question aloud if they ask, but do not help with the maths. If they are stuck, say "Try your best, or skip it", and move on. Write the answer in the box. Working on scrap paper is fine.

Warm-up

1 47 + 38 = ?

2 305 - 128 = ?

3 There are 4 packets. Each packet has 8 stickers. How many stickers in all?

4 24 sweets are shared equally among 3 children. How many sweets does each child get?

Main questions

5 Find the missing digit: 2 _ 7 + 358 = 645. What digit goes in the blank?

6 Mei has 3 times as many stickers as Ravi. Together they have 48 stickers. How many stickers does Mei have?

7 52 children sit in rows of 8. How many full rows are there, and how many children are left over?

8 Fill in the blank: 3/4 = ?/12

9 A rectangle is 9 cm long and 4 cm wide. What is its area? Write your answer with units.

10 A film starts at 2:45 p.m. and lasts 1 hour 50 minutes. At what time does it end?

Brain stretchers

11 I think of a number. I multiply it by 6, then subtract 4. My answer is 38. What is my number?

12 Use the digits 2, 5 and 8 to make 2-digit numbers. Each number must use two different digits. How many different numbers can you make?

FOR THE TUTOR OR PARENT

Answer key and scoring

Accept equivalent forms (for example $2.05 and 2.05). One mark per question.

Warm-up (4 marks)

#	Question	Answer	Watch for
1	47 + 38 = ?	**85**	Watch for forgetting to carry the 1 (gives 75).
2	305 - 128 = ?	**177**	Watch for borrowing across the zero; common error 223 (smaller digit minus bigger).
3	There are 4 packets. Each packet has 8 stickers. How many stickers in all?	**32**	Does the child know 8 x 4 or count/add in 8s? Note speed and strategy.
4	24 sweets are shared equally among 3 children. How many sweets does each child get?	**8**	Check the child shares equally (division), not subtracts 24 - 3.

Main questions (6 marks)

#	Question	Answer	Watch for
5	Find the missing digit: 2 _ 7 + 358 = 645. What digit goes in the blank?	**8**	Ask how they found it. Good: work from the ones (7 + 8 = 15, carry). Weak: guess and check only.
6	Mei has 3 times as many stickers as Ravi. Together they have 48 stickers. How many stickers does Mei have?	**36**	Look for a bar model: Ravi 1 unit, Mei 3 units, 4 units = 48, 1 unit = 12. Common error: 16 (that is 48 / 3) or 12 (Ravi's).
7	52 children sit in rows of 8. How many full rows are there, and how many children are left over?	**6 full rows, 4 left over**	Child should use 8 x 6 = 48. Error: 7 rows from rounding up, or giving only 6.
8	Fill in the blank: 3/4 = ?/12	**9**	Does the child multiply top and bottom by 3? Common error: adding 8 to both (11/12).
9	A rectangle is 9 cm long and 4 cm wide. What is its area? Write your answer with units.	**36 square cm**	Common error: giving perimeter 26 cm or writing 36 cm without 'square'. Ask what area means.
10	A film starts at 2:45 p.m. and lasts 1 hour 50 minutes. At what time does it end?	**4:35 p.m.**	Look for jumping: +1 hour = 3:45, +15 min = 4:00, +35 min = 4:35. Common error: 3:95 or 4:15.

Brain stretchers (2 marks)

#	Question	Answer	Watch for
11	I think of a number. I multiply it by 6, then subtract 4. My answer is 38. What is my number?	**7**	Strong reasoners work backwards: 38 + 4 = 42, 42 / 6 = 7. Others guess and check, which is also fine.
12	Use the digits 2, 5 and 8 to make 2-digit numbers. Each number must use two different digits. How many different numbers can you make?	**6**	Are they listing in order (25, 28, 52, 58, 82, 85)? Common errors: 3 or 9 (allowing repeats).

YOUR CHILD'S RESULT

Which level fits?

Add up the marks, then find your row. This is a guide. You make the final choice, and you can change level free of charge before lesson 1.

Warm-up marks: _____ / 4 **Main + stretch marks:** _____ / 8

Main + stretch	What it suggests
6 to 8	Join **Ages 8-10 (P3)**. Your child has the foundations and will be well challenged. If they scored 8 of 8 and all 4 warm-up marks, you can also try the Ages 9-11 (P4) check.
4 or 5	Join **Ages 8-10 (P3)**, and expect some support. If the warm-up score is 2 or less, Ages 7-9 (P2) may be the more comfortable start.
0 to 3	We suggest **Ages 7-9 (P2)**. The ideas build step by step, and a confident start matters more than a fast one.

Strong?
Choose the level that fits the score, not the school year. A child who finds this easy will enjoy the challenge questions in class.

Not yet?
That is useful to know, not a failure. Starting one level down means more success and more joy in class.

Skills covered

Addition and subtraction with regrouping; Missing digits; Bar models; Division with remainders; Equivalent fractions; Area of rectangles; Time; Working backwards and systematic listing

Next step. Send the score and your preferred level to us when you enrol. Prices are the same at every level, and your tutor can talk it through first. Small groups of up to 8, live online.